

Gabriel Macias de Oliveira

✉ macias.oliveira.gabriel@gmail.com [in Gabriel Macias de Oliveira](#) [🌐 Gabriel Macias](#) [🏠 Homepage](#)

Post-masters ML researcher with published biomedical research and production ML experience across financial, industrial, and biometric systems. Currently a research assistant at HHU Düsseldorf working on generative models for medical imaging. Interested in the foundations of generative modeling for scientific domains, particularly efficiency, controllability, and safety as the technical problems that determine whether these systems can be trusted in real applications. Seeking PhD positions and research collaborations.

Education

- RWTH Aachen**, MSc. Electrical and Computer Engineering 10/2022 - 11/2024
- Top 2 University in Germany in Engineering. GPA: 1.1 / 1.0 with distinction, top 1% of class.
 - Part of the highly selective T.I.M.E. Double Degree program (< 5% approved per year).
 - Member of the Dean's List 2022, top 5% of year.
 - **Master Thesis:** Enhancing Chest X-Ray Zero-Shot Classification via Extended Vision-Language Models.
- University of São Paulo**, BSc. Electrical and Computer Engineering 01/2019 - 08/2025
- Top 1 University of Latin America in Engineering. GPA: 9.2 / 10.0, ranked 1st out of 243 in class. Ranked 8th in National Entrance Exam.

Publications

- [1] Scazufca, M., Nakamura, C.A., Macias, G. et al. Self-help mobile messaging intervention for depression among older adults in resource-limited settings: a randomized controlled trial. *Nature Medicine* 30, 1127–1133 (2024).
- Designed and implemented the cohort selection pipeline from state health databases, identifying 700 eligible participants from 300k+ records under the trial's inclusion criteria.
- [2] Nakamura, C.A., Scazufca, M., Macias, G. et al. A self-help mobile messaging intervention to improve subthreshold depressive symptoms among older adults in a socioeconomically deprived region of Brazil (PRODIGITAL): a pragmatic, two-arm randomised controlled trial. *The Lancet Regional Health – Americas* 39 (2024).
- Built the data infrastructure tracking 10k+ weekly participant interactions that fed the trial's primary outcome analysis.

Work Experience

- Research Assistant**, HHU Düsseldorf 02/2026 - Present
- Built an end-to-end pipeline for automated acquisition and analysis of fungal infections in plant cell microscopy.
 - Developed a synthetic data generation pipeline using generative models for neuron spine segmentation in confocal microscopy, addressing dense annotation scarcity.
 - Implemented and trained foundational generative architectures from scratch ([DDPM](#), [DiT](#), [VQGAN](#), [LLMs](#)) to study their scaling behaviors and sampling efficiency.
- Machine Learning Engineer**, BTG Pactual 08/2025 - 01/2026
- Architected and deployed highly concurrent, autonomous agent systems, optimizing inference latency to handle massive client-interaction scaling (10k+ sessions/month).
 - Engineered a high-throughput OCR pipeline, optimizing model deployment, to automate tax calculation from foreign currency invoices, achieving a 40× reduction in processing time while maintaining 98% accuracy.
 - Led the technical development of an internal model monitoring framework, ensuring robust evaluation and safety constraints across 50+ deployed machine learning services.
- Machine Learning Engineering Intern**, QI Tech 11/2024 - 02/2025
- Developed and optimized a biometric spoof-detection algorithm, directly improving model robustness by 5% and satisfying strict ISO 30107-3 Level 2 technical requirements.
 - Designed and implemented an end-to-end MLOps pipeline for automated training, evaluation, deployment, and rollback of risk models.
- Software Engineering Intern**, Robert Bosch GmbH 10/2023 - 10/2024
- Engineered a RAG framework for automatic mechanical requirements extraction from image and text, reducing the processing time by 10× per document.
 - Implemented and optimized CI/CD pipelines for MEMS sensor software.

Teaching Experience

Teaching Assistant HHU Düsseldorf: Image Processing and Analysis; Advanced ML for Medical Data (2026) · RWTH Aachen: Digital Image Processing (2023) · University of São Paulo: Differential and Integral Calculus I & II (2020)

Awards and Scholarships

Springorum Commemorative Coin (2024) · Academic Merit Scholarship for International Exchanges, < 1% (2022-2023) · FAPESP Young Researchers Scholarship (2021) · Mathematics Institute Teaching Scholarship, < 0.1% (2020) · Brazilian Ministry of Health Grant (2020) · State Medalist, Physics/Math/Chem Olympiads (2014-2018)